Quantum Correlations Cannot Be Reproduced with a Finite Number of Measurements in Any No-Signaling Theory

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arXiv: 2408.08347

Main Result:

See Title

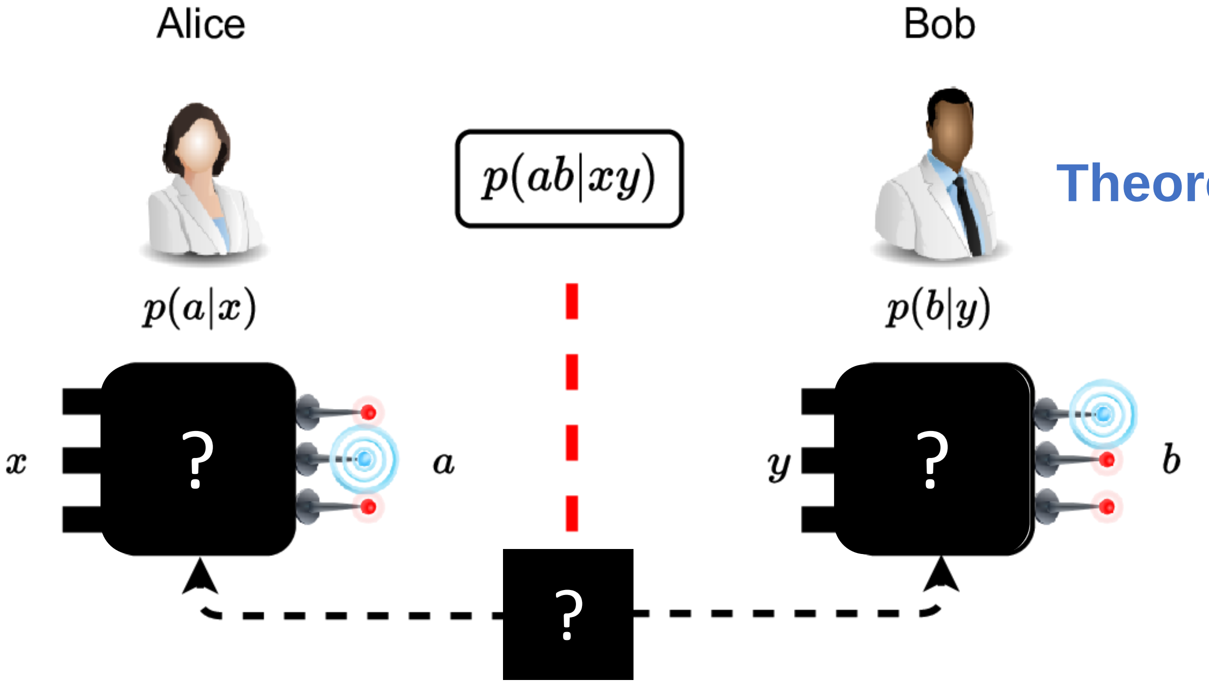
PhiQus

Thanks for organizing YQIS!

Correlation Scenarios

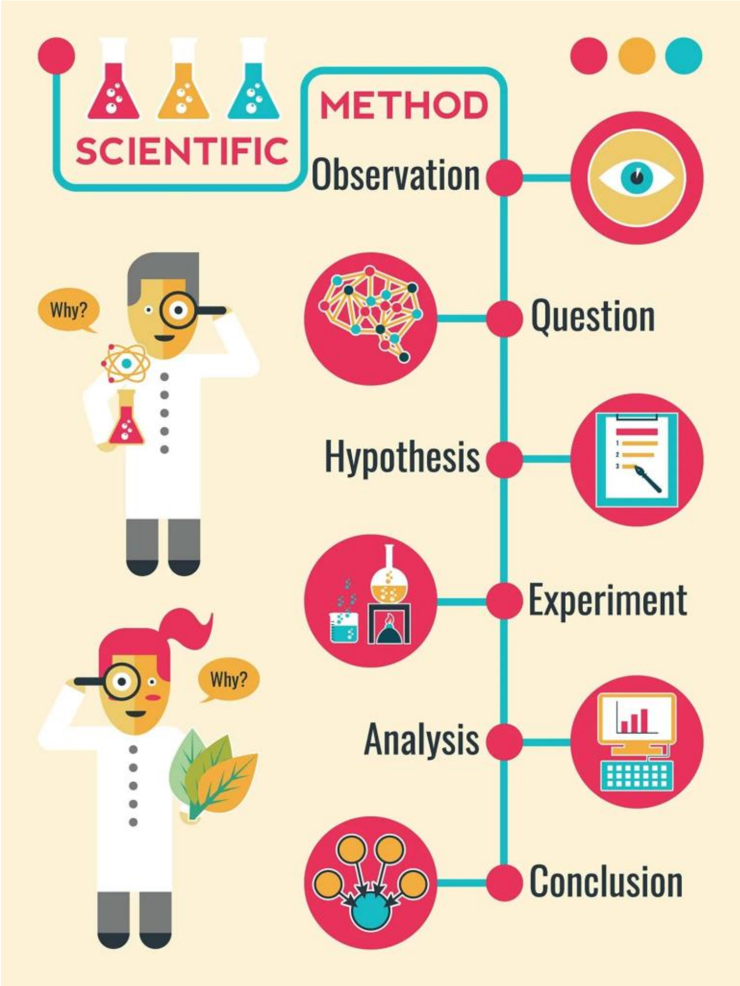
Bell experiment

$$p(ab|xy) \neq p(a|x)p(b|y)$$



How to explain the correlations?

Theoretical model of nature



https://www.simplypsychology.org/steps-of-the-scientific-method.html

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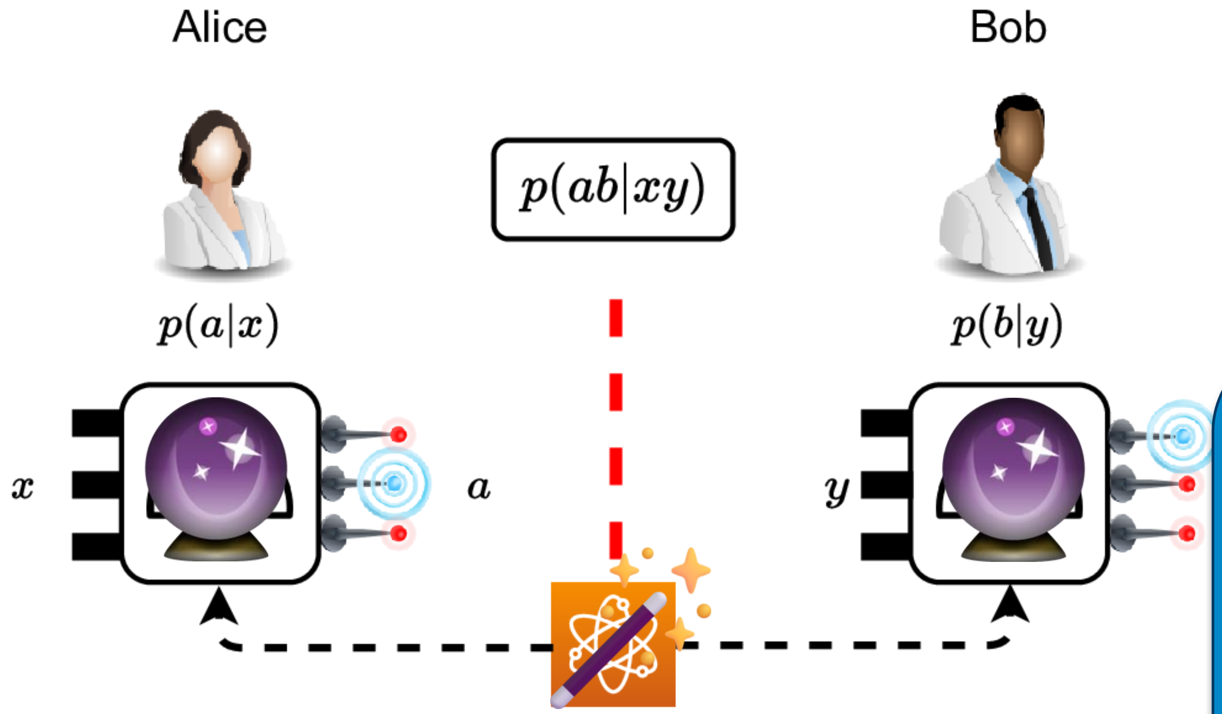
Explaining Correlations



List of possible explantations

1. ~~Classical explanations~~
2. Quantum Theory (QT)
3. More exotic theories

QT works well. Why actually?

Why are we not seeing the more exotic stuff?

Is there a more accurate description than QT?

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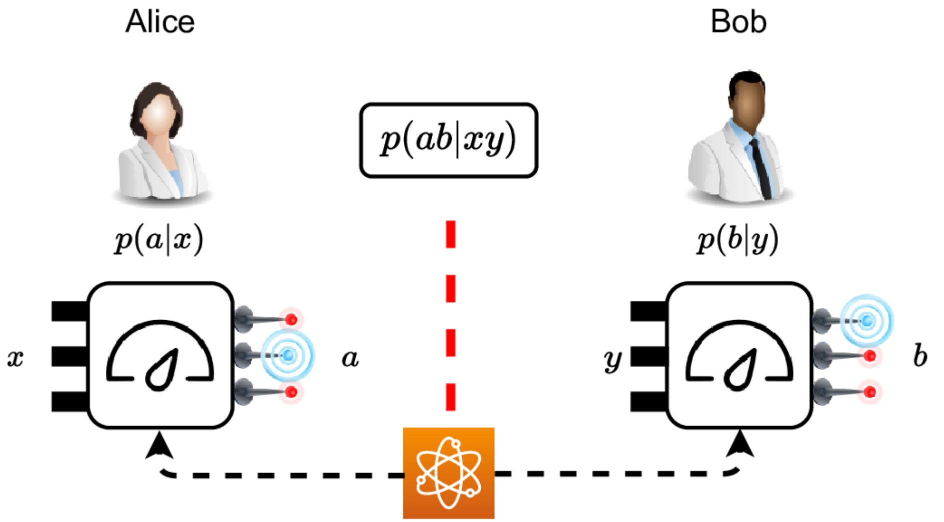
Quantum Theory

Quantum theory 101:

1. **States in complex Hilbert space** $|\Psi\rangle \in \mathcal{H}$ $\longrightarrow$ $\rho = \sum_i p_i |\Psi_i\rangle\langle\Psi_i| \iff \rho \geq 0, \mathrm{Tr}[\rho] = 1$
2. **Evolution of systems** $\rho \longmapsto \rho' = U\rho U^\dagger$
3. **Measurements** $\{M_a\}_a$ s.t. $0 \leq M_a \leq \mathbb{1}, \sum_a M_a = \mathbb{1} \longrightarrow p(a) = \mathrm{Tr}[M_a \rho]$
4. **Composite systems** $\mathcal{H}_{ST} := \mathcal{H}_S \otimes \mathcal{H}_T$ $\longrightarrow$ $\rho_{ST} := \rho_S \otimes \rho_T$



Quantum correlations: $\mathbf{p(ab|xy) = \mathrm{Tr}[(A_{a|x} \otimes B_{b|y})\rho]}$

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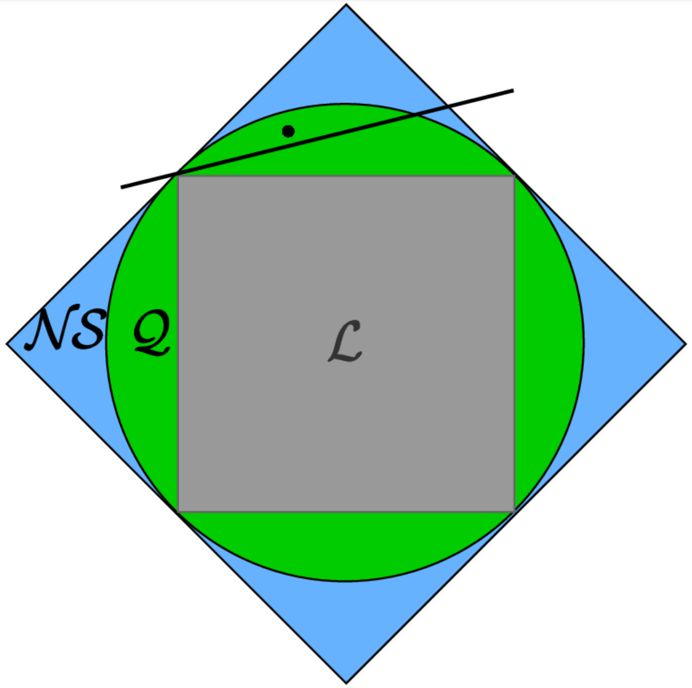
No-signaling as Physical Principle

Which physical principles constrain correlations?

No-signaling $\longrightarrow$ Causality

$$p(a|x) = p(a|xy) = \sum_b p(ab|xy)$$

$$p(b|y) = p(b|xy) = \sum_a p(ab|xy)$$



CHSH inequality: $\langle A_1 B_1\rangle + \langle A_1 B_2\rangle + \langle A_2 B_1\rangle - \langle A_2 B_2\rangle \overset{\mathcal{L}}{\leq} 2 \overset{\mathcal{Q}}{\leq} 2\sqrt{2} \overset{\mathcal{NS}}{\leq} 4$

with $\langle A_x B_y\rangle = p(a=b|xy) - p(a\neq b|xy)$

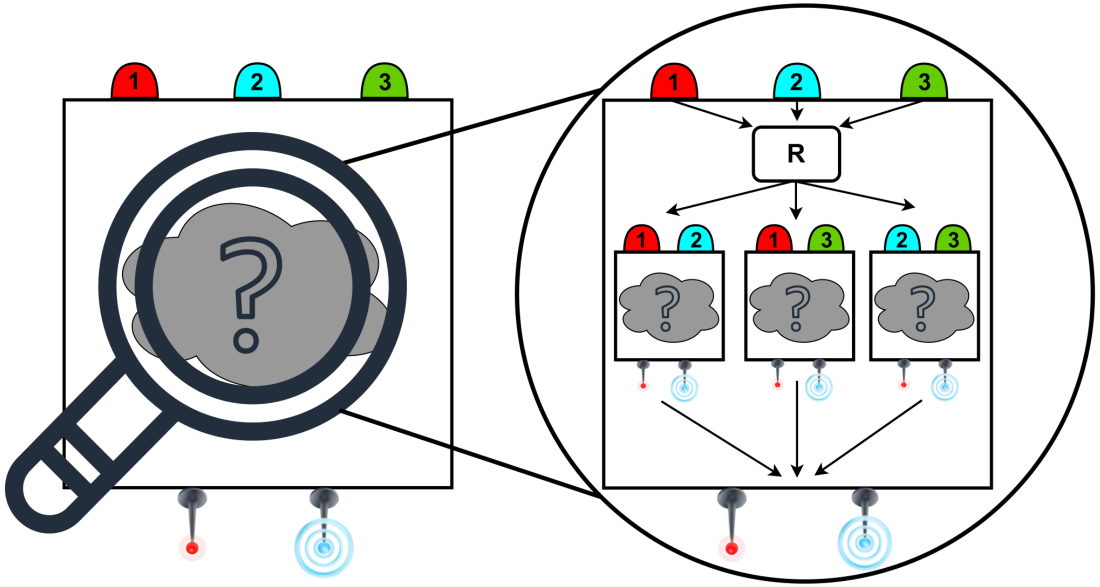
No-signaling theories are generally stronger than QT

Quantum correlations cannot be reproduced with a finite number of measurements in any no-signaling theory

Goal

Show that QT cannot be reproduced by any no-signaling theory with a finite number of measurements

1. What does it mean to perform genuinely n measurements?
2. How to prove it device-independently?
3. How to prove it theory-independently?



Measurement incompatibility

and its relation to nonlocality

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Measurement Incompatiblity

QM Lecture

Heisenberg-Robertson uncertainty relation

$$\Delta_\psi A \cdot \Delta_\psi B \geq \frac{1}{2}|\langle [A,B]\rangle_\psi|$$

Noncommutativity of observables

$$[A, B] = AB - BA \neq 0$$

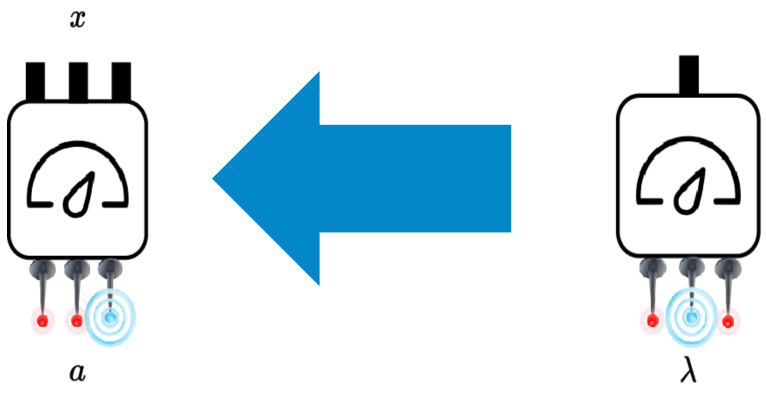
Observables $\leftrightarrow$ Projective measurements

$$A = \sum_a \mu_a P_a \longrightarrow \{P_a\}$$
$$B = \sum_b \nu_b Q_b \longrightarrow \{Q_b\}$$

Quantum Information

Joint measurability

$$M_{a|x} = \sum_\lambda p(a|x,\lambda) G_\lambda \iff M_{a|x} = \sum_{\vec{a}\setminus a_x} G_{\vec{a}}$$



JM implies LHV: $A_{a|x} = \sum_\lambda p(a|x,\lambda) G_\lambda \longrightarrow p(ab|xy) = \mathrm{Tr}[(A_{a|x} \otimes B_{b|y})\rho] = \underbrace{\sum_\lambda p_\lambda p(a|x,\lambda) p(b|y,\lambda)}_{\text{LHV}}$

$$\boxed{\sum_\lambda p(a|x\lambda)p(\lambda b|y) = \sum_\lambda p(a|x\lambda)p(b|y\lambda)p(\lambda|y) = \sum_\lambda p(\lambda)p(a|x\lambda)p(b|y\lambda) \quad \text{just using no-signaling}}$$

PRL **103**, 230402 (2009)

Quantum correlations cannot be reproduced with a finite number of measurements in any no-signaling theory

Incompatibility of two Measurements

Noisy Pauli measurements

$$M^{\lambda}_{\pm|1} = \frac{1}{2}(\mathbb{1} \pm \lambda\sigma_x) \qquad M^{\lambda}_{\pm|2} = \frac{1}{2}(\mathbb{1} \pm \lambda\sigma_z)$$

Parent POVM

$$\{G^{\lambda}_{s,t} = \tfrac{1}{4}[\mathbb{1} + \lambda(s\sigma_x + t\sigma_z)]\}_{s,t=\pm 1}$$

Marginals

$$M^{\lambda}_{\pm|1} = \textstyle\sum_t G^{\lambda}_{s,t} \qquad M^{\lambda}_{\pm|2} = \textstyle\sum_s G^{\lambda}_{s,t}$$

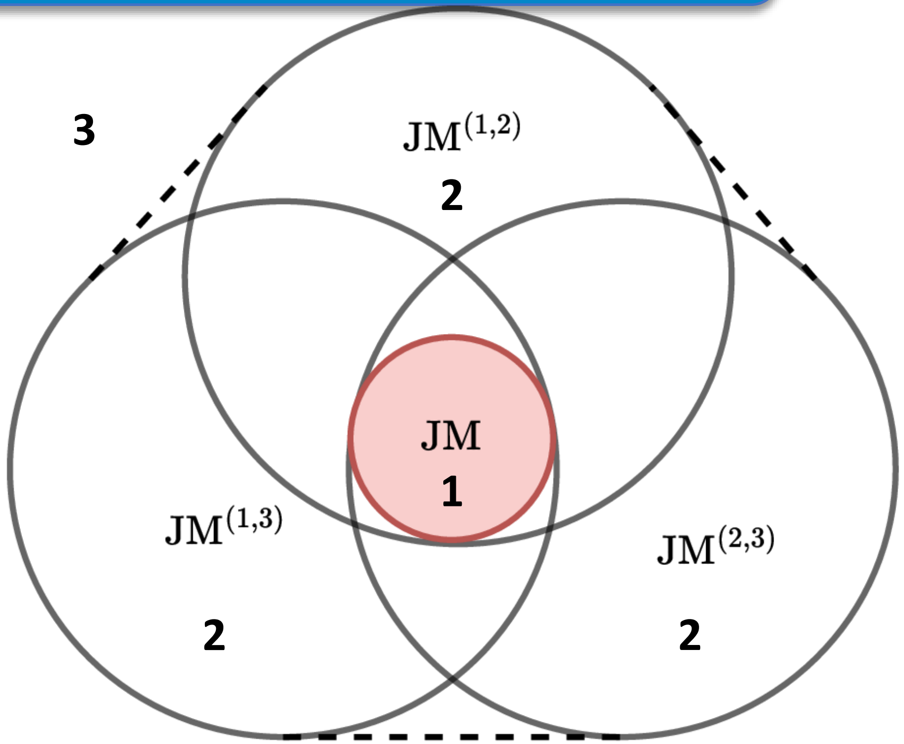
POVM

Completeness: ✔ Positivity: $\lambda \in [0, 1/\sqrt{2}]$

- Jointly measurable (JM) for $\lambda \in [0, 1/\sqrt{2}]$
- Jointly measurable but noncommunting

Incompatibility of multiple (≡ three) Measurements

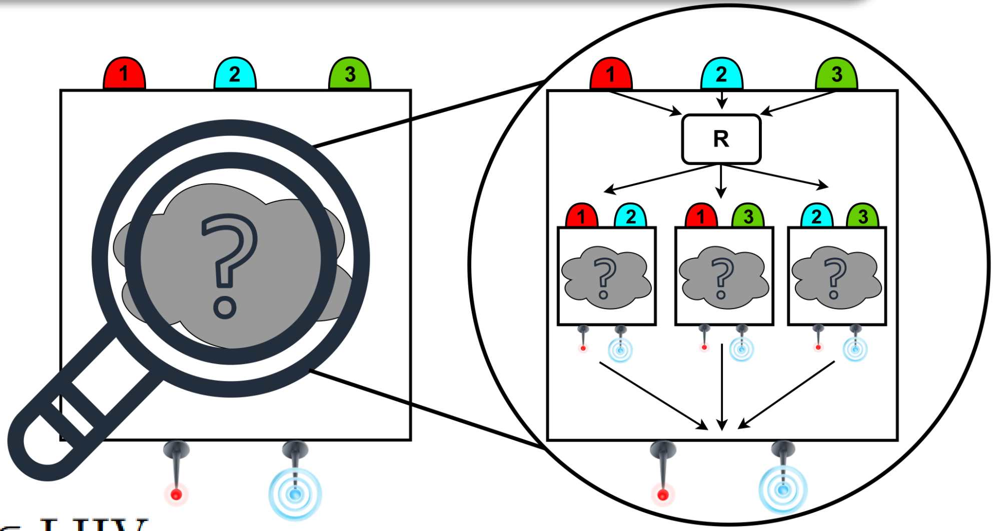
- JM: $M_{a|x} = \sum_\lambda p(a|x,\lambda) G_\lambda$
- JM pairs: $\mathrm{JM}^{(1,2)}, \mathrm{JM}^{(1,3)}, \mathrm{JM}^{(2,3)}$
- Pairwise: $\mathrm{JM}^{\mathrm{pair}} := \mathrm{JM}^{(1,2)} \cap \mathrm{JM}^{(1,3)} \cap \mathrm{JM}^{(2,3)}$
- Genuine triplewise: $\mathcal{M}_{(1,2,3)} \notin \mathrm{Conv}(\mathrm{JM}^{(1,2)}, \mathrm{JM}^{(1,3)}, \mathrm{JM}^{(2,3)})$
- Noisy Pauli X,Y, Z measurements: $M^{\eta}_{\pm|i} = \frac{1}{2}(\mathbb{1} \pm \eta\sigma_i)$
  - JM: $\eta \leq 1/\sqrt{3} \approx 0.58$
  - Pairwise: $\eta \leq 1/\sqrt{2} \approx 0.71$
  - Genuine triplewise incompatible: $\eta \geq \frac{\sqrt{2}+1}{3} \approx 0.8$



We can now count measurements!

PRL 123, 180401 (2019)

Device and Theory-Independent Counting of Measurements



Probabilistically faking 3 measurements

$$A_{a|x} = \sum_{(s,t)\in\{(1,2),(1,3),(2,3)\}} p_{(s,t)} F^{(s,t)}_{a|x} \quad \textbf{with} \quad \mathcal{F}^{(s,t)} \in \mathrm{JM}^{(s,t)}$$

Device-Independently

$$p(ab|xy) = \sum_{(s,t)\in\{(1,2),(1,3),(2,3)\}} p_{(s,t)} p^{(s,t)}(ab|xy) \quad \textbf{with} \quad p^{(s,t)}(ab|xy)|_{x\in(s,t)} \in \mathrm{LHV}$$

Theory-Independently

Assume only NS and not QT

Generally

$$A_{a|x} = \sum_{(s,t\neq s)\in\mathcal{T}} p_{(s,t)} F^{(s,t)}_{a|x} \longrightarrow p(ab|xy) = \sum_{(s,t\neq s)\in\mathcal{T}} p_{(s,t)} p^{(s,t)}(ab|xy)$$

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Certifying 3 Measurements Theory-independently

Choose convinient representation → **Collins-Gisin:**

$$\begin{array}{c||cc} & P(A_1) & P(A_2) \\ \hline P(B_1) & P(A_1B_1) & P(A_2B_1) \\ P(B_2) & P(A_1B_2) & P(A_2B_2) \end{array}$$

$P(A_1) := p_A(a=0|x=1), P(A_1B_1) := p(a=0,b=0|x=1,y=1)$

Example: $\text{CHSH} = \begin{array}{c||cc} & -1 & 0 \\ \hline -1 & 1 & 1 \\ 0 & 1 & -1 \end{array} \leq 0$

$P(A_k) \geq P(A_kB_j)$ and $P(B_k) \geq P(A_jB_k)$

From no-signaling

$$M_{3322} = \begin{array}{c||ccc} & -2 & 0 & 0 \\ \hline -2 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ 0 & 1 & -1 & 0 \end{array} \leq 0$$

Violation of M3322 implies violation of:

$$M_{3322}^{(1,2)} = \begin{array}{c||ccc} & -1 & 0 & 0 \\ \hline -1 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{array} \leq 0 \qquad M_{3322}^{(1,3)} = \begin{array}{c||ccc} & -1 & 0 & 0 \\ \hline -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{array} \leq 0 \qquad M_{3322}^{(2,3)} = \begin{array}{c||ccc} & -1 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{array} \leq 0$$

All pairs have to be incompatible!

Convex combinations cannot help!

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A No-Go Theorem

For any n ≥ 2 the violation of the Mnn22 inequality certifies genuine n measurements in any NS theory

$$M_{nn22} = \begin{array}{c||cccccc} & -(n-1) & 0 & 0 & \cdots & 0 & 0 \\ \hline -(n-1) & 1 & 1 & 1 & \cdots & 1 & 1 \\ -(n-2) & 1 & 1 & 1 & \cdots & 1 & -1 \\ -(n-3) & 1 & 1 & 1 & \cdots & -1 & 0 \\ \vdots & \vdots & & & & & \vdots \\ -1 & 1 & 1 & -1 & \cdots & 0 & 0 \\ 0 & 1 & -1 & 0 & \cdots & 0 & 0 \end{array} \leq 0$$

Proof steps

1. Violation of Mnn22 implies violation of M(n-1)(n-1)22
2. Bob's first measurement is pairwise incompatible with any other measurement
3. Use it for induction type proof
4. Violation certifies incompatibility between all measurement pairs

Quantum violation

$$|\Psi_\epsilon\rangle = \sqrt{\frac{1-\epsilon^2}{n-1}} \sum_{k=1}^{n-1} |kk\rangle + \epsilon|nn\rangle \quad \textbf{with} \quad \epsilon^2 = \frac{1-q_0^2}{1+[(n-1)^2-1]q_0^2} \text{ and } 0 \leq q_0 \leq 1$$

Projective measurements paramatrized by q_0

$$\longrightarrow \quad M_{nn22} = \epsilon^2(q_0^2 n - (n-1)) > 0 \quad \textbf{for} \quad \sqrt{\frac{n-1}{n}} < q_0 < 1$$

Tailored for Inn22 inequality

Phys. Rev. Lett. 104, 060401 (2010)

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Experimental Considerations

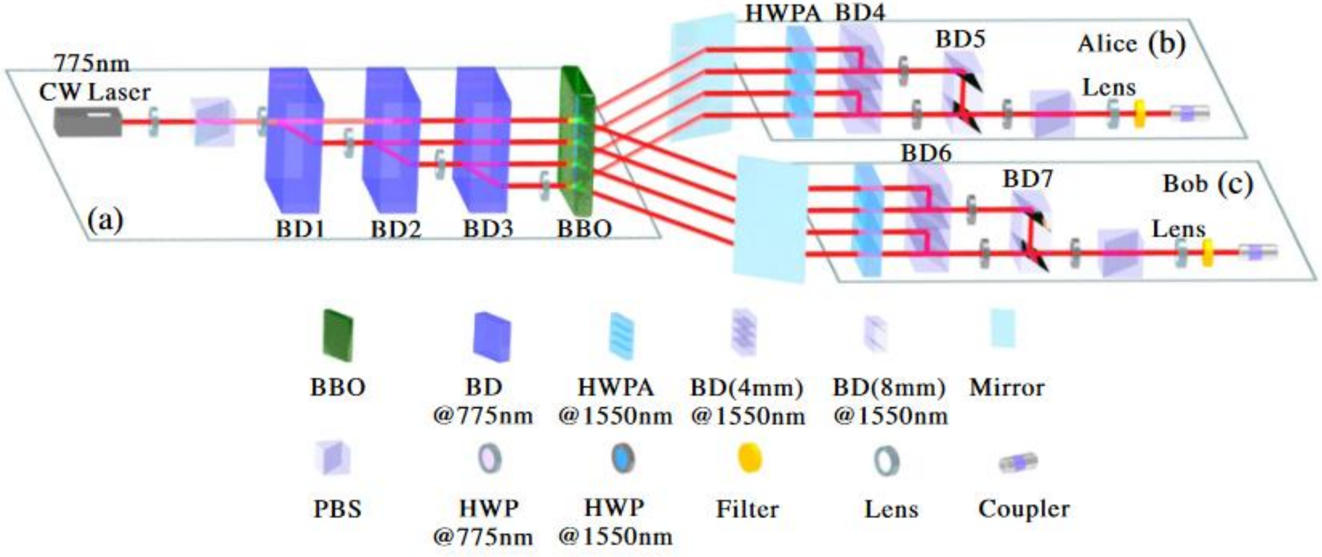
Can we falsify (n-1) measurements in an actual experiment? ⟶ Certainly not for all n…

Challenges

1. Low violations
   - n = 3: $M_{3322} \approx 0.0239$
   - n = 5: $M_{5522} \approx 0.0035$
2. High-dimensional entanglement
3. The measurements themselves?

State-of-the-Art

1. n = 3 violation has been achieved PRX **5,** 041052 (2015)
2. Similar experiments with d = 4 using path entanglement PRL **129**, 060402 (2022)



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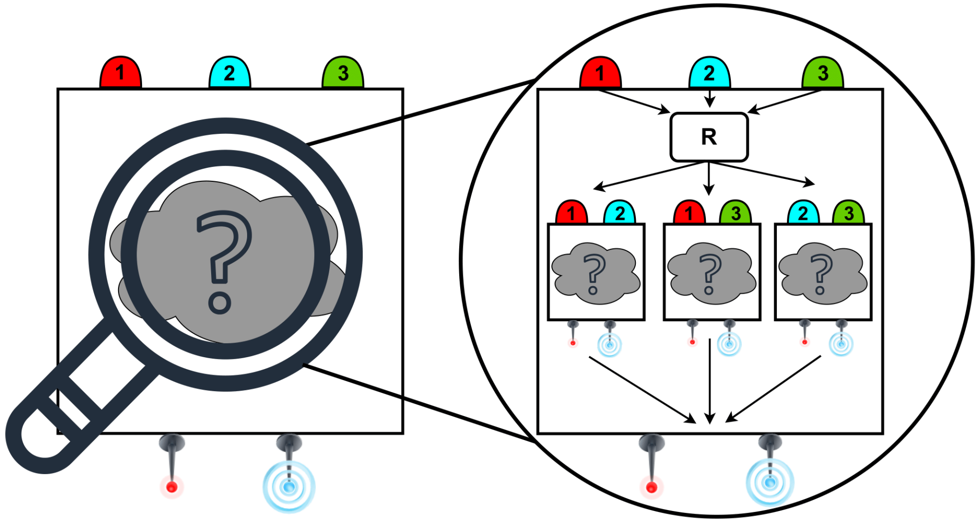
Summary

- Finite number of measurements in any NS-theory fail to reproduce predictions of quantum theory
- Bell experiment with 2 parties, 2 outcomes, genuinely n measurements

arXiv: 2408.08347

PRL **135**, 030201 (2025)



$$M_{nn22} = \begin{array}{c||cccccc} & -(n-1) & 0 & 0 & \cdots & 0 & 0 \\ \hline\hline -(n-1) & 1 & 1 & 1 & \cdots & 1 & 1 \\ -(n-2) & 1 & 1 & 1 & \cdots & 1 & -1 \\ -(n-3) & 1 & 1 & 1 & \cdots & -1 & 0 \\ \vdots & \vdots & & & & & \vdots \\ -1 & 1 & 1 & -1 & \cdots & 0 & 0 \\ 0 & 1 & -1 & 0 & \cdots & 0 & 0 \end{array} \leq 0$$

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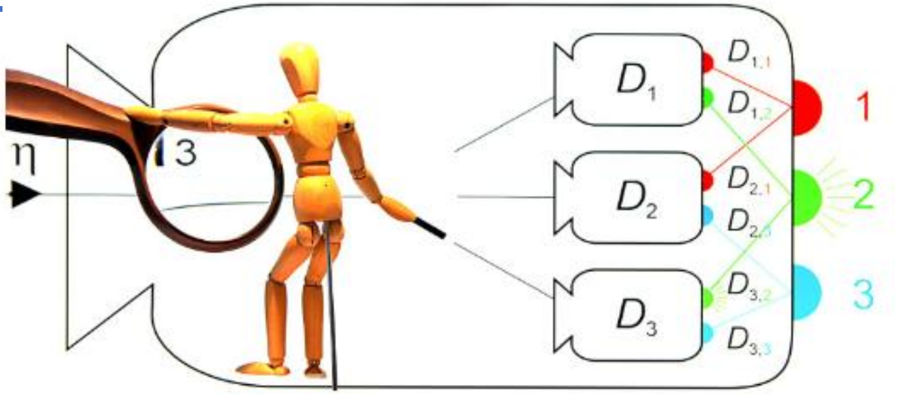
The End

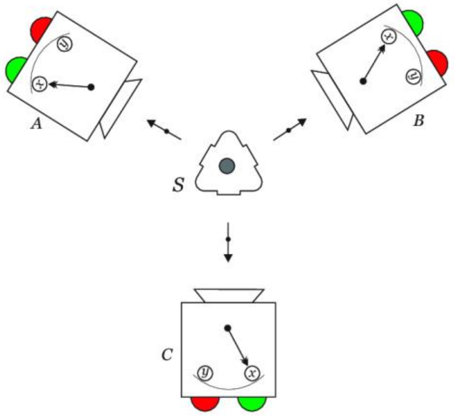
Well…

Restricted Causal Theories vs. QT

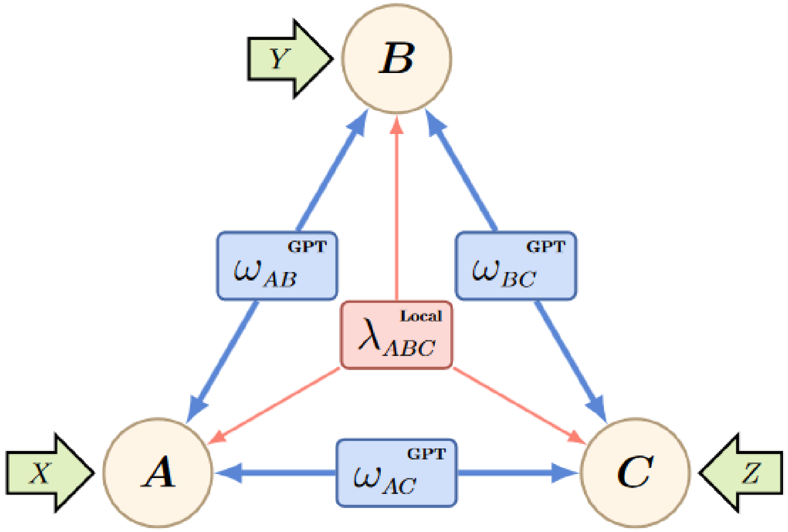
Consider a restriction to general causal theories

- **(n-1) measurement outcomes** PRL **117**, 150401 (2016)
- **(n-1) entangled systems** PRL **88**, 170405 (2002)
- **(n-1) partite sources** PRL **127**, 200401 (2021)





arxiv:1909.00862



QT is stronger than any (n-1) restricted causal theory for any n ≥ 2.

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Incompatibility and Nonlocality

JM implies LHV: $A_{a|x} = \sum_\lambda p(a|x,\lambda) G_\lambda \longrightarrow p(ab|xy) = \mathrm{Tr}[(A_{a|x} \otimes B_{b|y})\rho] = \underbrace{\sum_\lambda p_\lambda p(a|x,\lambda) p(b|y,\lambda)}_{\text{LHV}}$

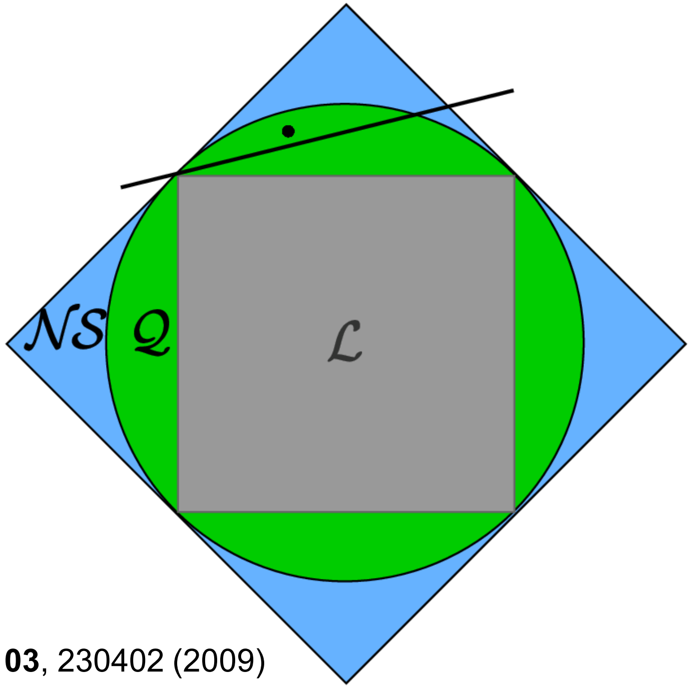
Bell inequality: $\sum_{a,b,x,y} C_{ab|xy} p(ab|xy) \overset{\mathcal{L}}{\leq} L \overset{\mathcal{Q}}{\leq} Q \overset{\mathcal{NS}}{\leq} \mathrm{NS}$

Note: Incompatibility not sufficient for nonlocality PRA **97**, 012129 (2018)

CHSH inequality: $\langle A_1 B_1 \rangle + \langle A_1 B_2 \rangle + \langle A_2 B_1 \rangle - \langle A_2 B_2 \rangle \overset{\mathcal{L}}{\leq} 2 \overset{\mathcal{Q}}{\leq} 2\sqrt{2} \overset{\mathcal{NS}}{\leq} 4$

with $\langle A_x B_y \rangle = p(a = b|xy) - p(a \neq b|xy)$

$\longrightarrow$ One-to-one correspondence! $\mathrm{CHSH} = \sqrt{4 + \|[A_1, A_2] \otimes [B_1, B_2]\|}$



$p(a_1, a_2) \implies p(a_1, a_2, b_1, b_2) \implies \mathrm{LHV}$ **just using no-signaling**

PRL **103**, 230402 (2009)

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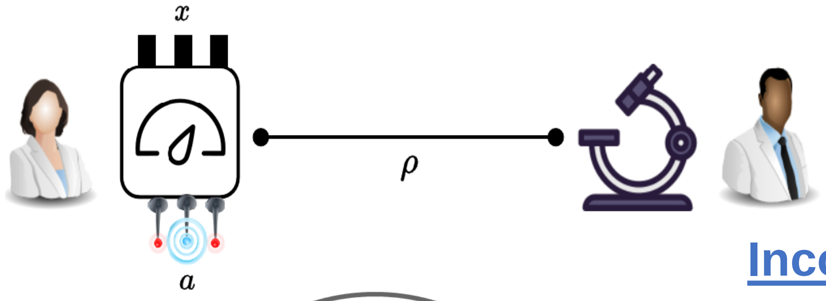
Related works

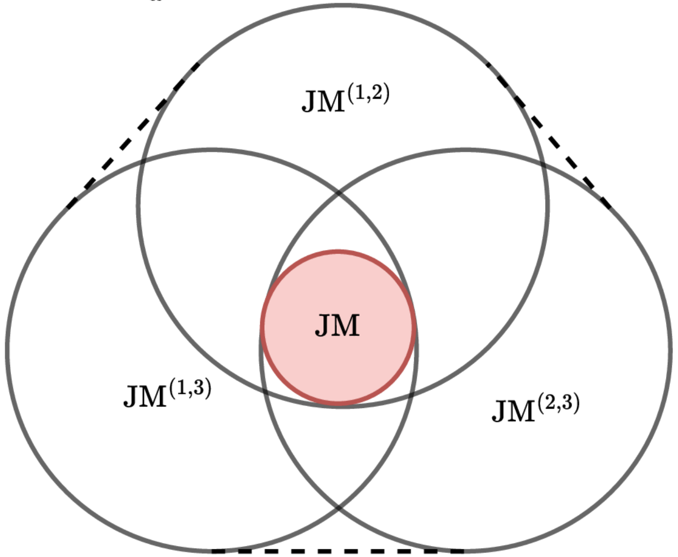
Semi-DI certification of number of measurements:

Semi-device-independent certification of the number of measurements

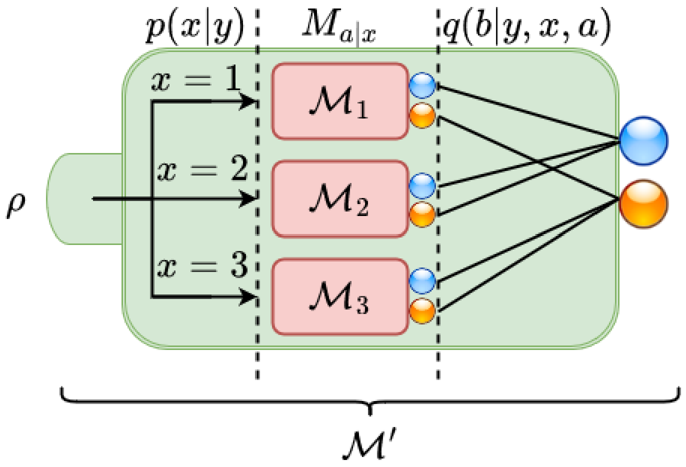
Isadora Veeren, Martin Plávala, Leevi Leppäjärvi, and Roope Uola
Phys. Rev. A **109**, 062203 – Published 3 June 2024

Steering





Incompatibility structures vs. Simulability



$$\xi : \mathcal{M} \xrightarrow{\xi} \mathcal{M}' \text{ s.t. } M'_{b|y} = \sum_x p(x|y) \sum_a q(b|y,x,a) M_{a|x}$$

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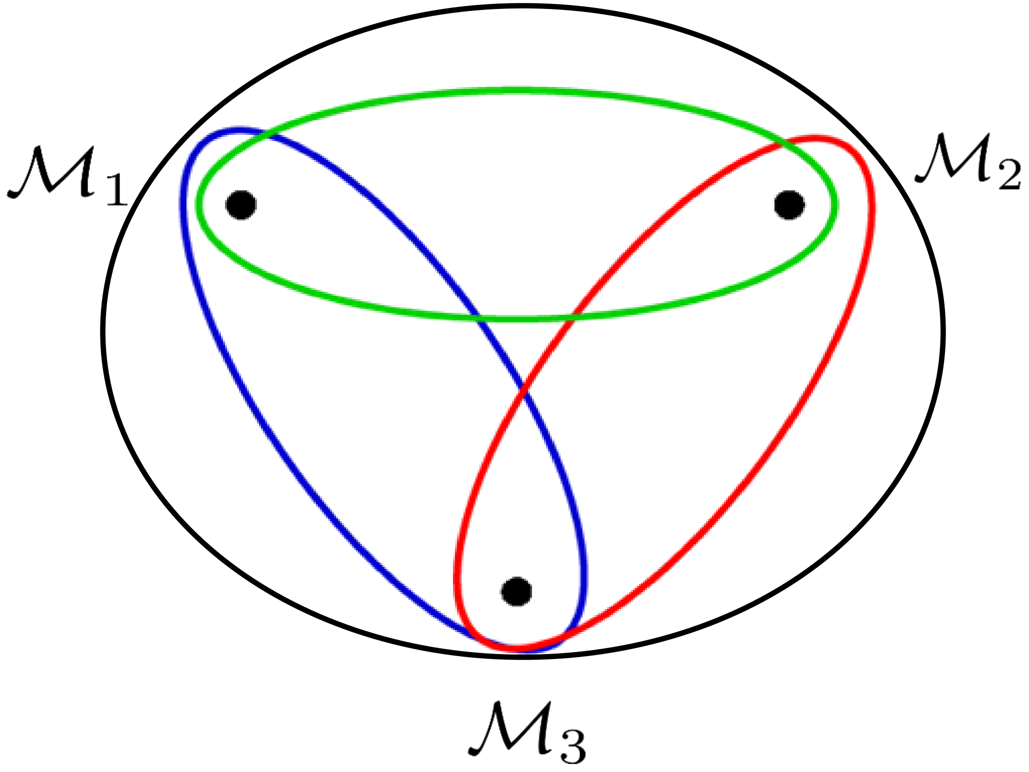
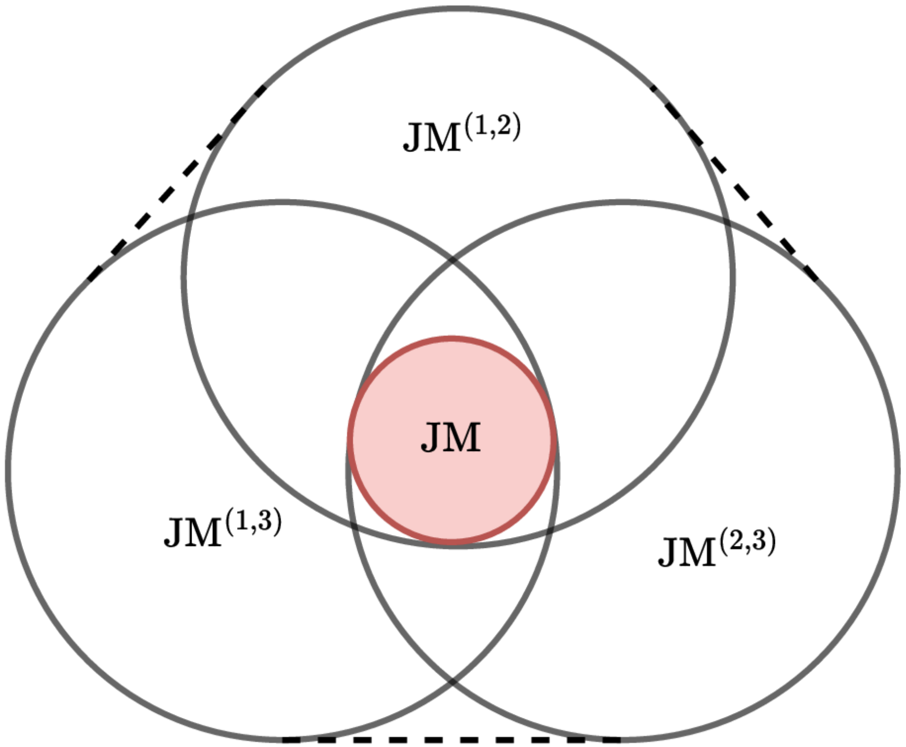
Beyond 2 Measurements





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